Lab 1, task #2 (Friday, April 27th 2026)

A company is planning to purchase a new laptop for its design department. There are six laptop models: A - F. The decision which laptop model to choose will be based on the following decision criteria:

1. **Price (PLN)** – lower is better *(cost criterion),*
2. **Performance (CPU benchmark score)** – higher is better *(benefit criterion),*
3. **Battery life (hours)** – higher is better *(benefit criterion),*
4. **Weight (kg)** – lower is better *(cost criterion),*
5. **Storage capacity (GB)** – higher is better *(benefit criterion).*

The relative importance of the criteria is described in linguistic terms (assumptions regarding the importance of particular criteria) is as follows:

- **Performance** is more important than **Price;**
- **Price** is more important than **Storage;**
- **Storage** is slightly more important than **Battery;**
- **Battery** is slightly more important than **Weight.**

The following table contains the specifications and benchmark data (decision matrix):

Laptop	Price (PLN)	CPU Performance (PassMark score)	Battery (h)	Weight (kg)	Storage (GB)
A	4500	15500	6	2.0	512
B	4000	14200	7	2.2	256
C	4800	17200	5	1.8	1024
D	4300	13800	8	2.5	512
E	4700	16000	6	2.1	512
F	4200	14500	9	2.0	256

P1: Which of the abovementioned laptop models would be the most appropriate for the design department assuming the importance of the criteria described above?

For given decision matrix and assumptions regarding the importance of particular criteria, do the following tasks (for total **max. 8 points**).

1. Determine criteria weight vector (w) by using AHP approach (construct binary comparison matrix and calculate its eigenvector) – **max. 2 points**.
2. Solve the P1 problem by using the calculated weight vector (w) and SAW method (hybrid SAW-AHP approach) – **max. 2 points**.
3. Solve the P1 problem by using the calculated weight vector (w) and TOPSIS method (hybrid TOPSIS-AHP approach) – **max. 3 points**.
4. Compare the resulting ranking vectors for P1 from points 2. and 3. , discuss possible reasons for the similarities or differences – **max. 1 point**.

It is recommended to solve the tasks above using a spreadsheet software (MS Excel, Libre Office Calc). However, it is also possible to solve these tasks by writing code in one's own favourite programming language, provided that no framework, library or built-in functionality is used for SAW, AHP or TOPSIS calculations.